

Connecting the Dots in AMR Surveillance

An interpretable network framework for structure, ranking and prospective prioritisation

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Abstract

Routine antimicrobial-resistance (AMR) surveillance is usually read one pathogen–antibiotic pair at a time. This reveals how often resistance is observed, but not how each profile is positioned within the wider resistance landscape. We address this gap by adapting Fitness–Complexity, a framework developed to extract relational information from country–product networks. Fitness rewards broad profiles containing selective links, whereas complexity is high when a link is confined to high-fitness profiles. Starting from Pfizer ATLAS data accessed through the Vivli AMR Register, we normalise resistance fractions among sufficiently tested pairs using revealed comparative resistance (RCA). An RCA-positive link identifies a pair whose resistance is at least as high as expected from the overall profiles of that pathogen and antibiotic within the same dataset.

The resulting pathogen–antibiotic network is not organised as the single nested diversification ladder characteristic of country–product trade. Instead, it is block-nested, as observed in firm–product systems: narrower pathogen profiles are approximately contained within broader profiles only inside data-driven modules. This distinction is essential. Global Fitness–Complexity scores collapse onto a few nodes, whereas scores computed within modules provide informative local rankings. Antibiotic complexity increases across the WHO Access, Watch and Reserve stewardship tiers, and pathogen fitness is higher for ESKAPEE taxa, defined here as the ESKAPE group plus *Escherichia coli*; neither reference was used to construct the network.

The same topology also predicts future comparative-resistance signals. A parameter-free exposure score ranks currently RCA-negative pairs from the RCA-positive links already surrounding them in antibiotic or pathogen relatedness networks. In prospective tests using past data only, the antibiotic-network score achieved a pooled AUC of 0.71, and remained stable over forecast horizons of one to five years. The framework therefore turns routine counts into an auditable early-warning ranking: every prioritised pair can be traced to observed resistance fractions, comparative links and explicit network neighbours, without fitting a black-box classifier.

Objectives

We pursued three linked objectives: (i) construct an interpretable pathogen–antibiotic network that identifies resistance associations that are comparatively prominent within the analysed surveillance dataset; (ii) characterise the network organisation suggested by previous Economic Complexity analyses and by the known heterogeneity of AMR, and determine the scale at which Fitness–Complexity yields interpretable pathogen and antibiotic rankings; and (iii) test prospectively whether exposure to existing network signals can prioritise pathogen–antibiotic pairs that become RCA-positive at one-year and longer forecast horizons.

Methods

Data, resistance fractions and revealed comparative resistance

The input contains annual species–antibiotic counts from the Pfizer ATLAS surveillance programme distributed through the Vivli AMR Register, spanning 394 species, 43 antibiotics and 2004–2024 [1]. For each analysis window, resistant and tested counts were pooled before applying the minimum-isolate threshold. A pair (p, a) was retained when at least $n_{\min} = 30$ isolates had been tested, and its resistance fraction was

$$r_{pa} = \frac{n_{pa}^R}{N_{pa}}$$

where intermediate isolates contribute to N_{pa} but not to n_{pa}^R . If the pair was absent or did not reach the threshold, r_{pa} was undefined. The complete workflow was also repeated at $n_{\min} = 10$ and 50, with all other settings unchanged.

Raw percentages are difficult to compare across a heterogeneous table. A high resistance fraction may reflect a pathogen that is broadly resistant, an antibiotic for which resistance is common across many pathogens, or a particularly prominent association between that pathogen and antibiotic. To isolate the third component from the first two, we adapted revealed comparative advantage from Economic Complexity [2, 3]. Let $\mathcal{O}$ be the sufficiently observed cells in an analysis window, $R_p = \sum_{a:(p,a) \in \mathcal{O}} r_{pa}$, $A_a = \sum_{p:(p,a) \in \mathcal{O}} r_{pa}$, and $T = \sum_{(p,a) \in \mathcal{O}} r_{pa}$. Then

$$\text{RCA}_{pa} = \frac{r_{pa}}{R_p A_a / T} = \frac{r_{pa} T}{R_p A_a}$$

Thus $\text{RCA}_{pa} \geq 1$ identifies a resistance fraction at least as large as the value implied by the pathogen and antibiotic marginals in the same dataset and time window; for example, $\text{RCA}_{pa} = 2$ denotes twice that marginal expectation. We call this an *RCA-positive* or *revealed comparative-resistance* link. It is a relative, within-dataset signal, not a clinical breakpoint or an

absolute biological label. The implemented binary matrix is

$$M_{pa} = \mathbf{1}\{\text{RCA}_{pa} \geq 1\}$$

Because comparisons with undefined RCA evaluate as false in the pipeline, $M_{pa} = 0$ combines sufficiently observed cells with $\text{RCA}_{pa} < 1$ and insufficiently observed cells. A zero therefore means “no RCA-positive link in the analysed surveillance universe”, not confirmed susceptibility.

Why measure nestedness?

Economic Complexity was developed for bipartite country–product networks. At the conceptual level, a link records that a country exports a product. Because countries may export small quantities of many products, empirical Fitness–Complexity analyses commonly use revealed comparative advantage to obtain the binary matrix on which the rankings are calculated: a link then retains a product that is important relative to that country’s export profile. Our construction follows the same normalisation logic, but with a different interpretation. Pathogens replace countries, antibiotics replace products, and a link denotes a comparatively prominent resistance association rather than simply that the pair was tested.

A bipartite matrix is *nested* when rows with few links tend to connect to subsets of the columns connected to rows with more links. In country–product trade, less diversified countries therefore export subsets of the products exported by more diversified countries, while products exported by few countries occur mainly in broad export baskets. Sorting rows and columns by degree produces one approximately triangular hierarchy. Firm–product networks may instead be *block-nested*: firms operate in different sectors, and subset hierarchies emerge mainly within those sectors [5].

We tested both possibilities in the AMR matrix. A pathogen’s degree is the number of antibiotics for which it has an RCA-positive link, and an antibiotic’s degree is the number of linked pathogens. Global nestedness would mean that narrow pathogen profiles are largely contained within broader profiles across the full dataset. In-block nestedness would mean that the same ordering is concentrated within data-driven pathogen–antibiotic modules. This distinction is consequential because Fitness–Complexity is most interpretable when applied to an asymmetric subset hierarchy: one global hierarchy supports one global ranking, whereas block-nested organisation calls for local rankings.

For two pathogen rows i, j , let k_i and k_j be their degrees and $O_{ij} = \sum_a M_{ia}M_{ja}$ their number of shared antibiotic neighbours. Under a degree-based expectation, $\langle O_{ij} \rangle = k_i k_j / A$, where A is the number of antibiotic columns. The score compares the observed overlap of a lower-degree profile with a higher-degree profile against this expectation; the analogous calculation is performed for antibiotic columns. The implemented in-block nestedness is [4, 5]

$$\mathcal{I}_R = \sum_{\substack{i,j \\ k_i > k_j, b_i = b_j}} \frac{O_{ij} - k_i k_j / A}{k_j (C_i - 1)} \quad , \quad \mathcal{I}_C = \sum_{\substack{a,c \\ k_a > k_c, b_a = b_c}} \frac{O_{ac} - k_a k_c / P}{k_c (C_a - 1)} \quad , \quad \mathcal{I} = \frac{2}{P + A} (\mathcal{I}_R + \mathcal{I}_C)$$

where b identifies a co-cluster and C its size. Global nestedness $\mathcal{N}$ uses the same statistic with all nodes assigned to one block.

The structural analysis used the pooled 2015–2024 matrix after removing rows and columns without positive links. Spectral co-clustering partitioned pathogen rows and antibiotic columns simultaneously into five matched blocks [6]. Significance was assessed with 100 Bernoulli matrices satisfying $\Pr(M_{pa}^{\text{null}} = 1) = \min(k_p k_a / E, 1)$, where $E = \sum_{pa} M_{pa}$. Empty nodes were removed and every null matrix was re-clustered. This null preserves degree heterogeneity in expectation before clipping, rather than fixing every degree exactly.

Local Fitness–Complexity rankings

Fitness–Complexity jointly assigns one score to each row and one to each column through an iterative feedback. In the country–product setting, a country’s fitness increases when it exports many products, particularly products with high complexity. Product complexity is high only when all of its exporters have high fitness: because the update is harmonic-like, even one low-fitness exporter strongly reduces the product’s score. Complexity therefore measures the selectivity of a complete link profile, not simply how rarely a product is exported.

Applied to the AMR network, *pathogen fitness* increases with the number and complexity of the antibiotics in a pathogen’s RCA-positive profile. A pathogen scores highly when its profile is broad and includes selective links. *Antibiotic complexity* is high when RCA-positive resistance to that antibiotic is concentrated among high-fitness pathogen profiles; a link to a low-fitness pathogen strongly reduces the score. Pathogen fitness is unrelated to evolutionary fitness, and antibiotic complexity is not a direct measure of drug potency, clinical efficacy or absolute resistance prevalence.

Starting from $F_p^{(0)} = Q_a^{(0)} = 1$, the pipeline iterates [3]

$$\tilde{F}_p^{(n)} = \sum_a M_{pa} Q_a^{(n-1)} \quad , \quad \tilde{Q}_a^{(n)} = \left(\sum_p \frac{M_{pa}}{F_p^{(n-1)}} \right)^{-1}$$

and normalises F and Q by their means at each step. Scores were calculated globally and separately within each co-cluster. Within-block scores were standardised before pooling, so they describe relative position within a module rather than an absolute scale shared across modules.

For external consistency, antibiotic complexity was compared across the WHO Access, Watch and Reserve (AWaRe) stewardship tiers [7]. Access antibiotics are generally preferred first- or second-choice treatments for common infections; Watch antibiotics have greater resistance-selection concerns; and Reserve antibiotics are intended for selected multidrug-resistant infections. Pathogen fitness was compared between taxa matching the ESKAPEE reference list—the classical ESKAPE group [8] extended here with *Escherichia coli*—and all other taxa. Neither reference was used to construct the network, define the modules or calculate the scores.

Relatedness networks and prospective prioritisation

For each origin year $t = 2019, \dots, 2023$, all quantities were rebuilt using only data from 2004 through t . The antibiotic projection counted pathogens linked to both antibiotics, and the pathogen projection counted antibiotics linked to both pathogens:

$$W_{ab}^A = \sum_p M_{pa}^{(t)} M_{pb}^{(t)} \quad , \quad W_{pq}^P = \sum_a M_{pa}^{(t)} M_{qa}^{(t)}$$

For a sufficiently observed pair (p, a) that was RCA-negative at t , antibiotic-network exposure was

$$\rho_{pa}^A = \frac{\sum_o W_{ao}^A M_{po}^{(t)}}{\sum_o W_{ao}^A}$$

the weighted fraction of antibiotic a 's network neighbourhood already RCA-positive for pathogen p . The analogous pathogen-network score was $\rho_{pa}^P = \sum_q W_{pq}^P M_{qa}^{(t)} / \sum_q W_{pq}^P$. The outcome was whether the same sufficiently tested pair had $\text{RCA}_{pa} \geq 1$ in year $t + 1$. The scores contain no fitted coefficients. Discrimination was measured by AUC against a baseline given by the antibiotic's current fraction of RCA-positive pathogen rows, with 200 node-label permutations used for significance comparisons.

To test persistence beyond the next year, we additionally fixed the history, network, exposure scores and initial candidate set at $t_0 = 2019$, and evaluated those unchanged scores against each target year from 2020 to 2024. At every horizon, evaluation was restricted to pairs sufficiently tested in the target year; the network was never refitted.

Results

The matrix contains several local hierarchies, not one global ladder

Pooling 2015–2024 yielded 1,932 sufficiently observed cells in the 139-by-40 structural universe; 991 were RCA-positive. Global nestedness was low ($\mathcal{N} = 0.0496$) and below the degree-corrected null (0.0844 ± 0.0081 , $z = -4.30$). In contrast, nestedness measured within the five co-clusters was $\mathcal{I}^* = 0.3855$, compared with 0.1453 ± 0.0229 under the null ($z = 10.51$). The ratio $\mathcal{I}^*/\mathcal{N} = 7.77$ shows that subset ordering was concentrated within modules rather than expressed as one hierarchy across the complete matrix.

The AMR network therefore resembles the block-nested organisation reported for firm–product systems more than the global diversification ladder of country–product trade. Pathogen and antibiotic profiles can be compared within related network contexts, whereas a single cross-module ordering forces distinct structures onto one axis. The blocks were inferred only from M ; no taxonomic, drug-class, AWaRe or ESKAPEE labels entered the clustering. Because insufficiently observed and sufficiently observed RCA-negative cells both enter M as zero, the inferred organisation reflects both comparative-resistance profiles and the testing and coverage pattern represented in ATLAS.

Block-conditioned scores recover clinically recognisable ordering

Applied to the complete matrix, the Fitness–Complexity iteration collapsed numerically onto one pathogen and one antibiotic: 138 of 139 global pathogen scores and 39 of 40 global antibiotic scores were below 10^{-10} after normalisation. This distribution does not provide a stable cross-module ranking. The collapse is consistent with the absence of one global nested hierarchy and with the nonlinear complexity update, which strongly penalises links to low-fitness pathogens. Computing the scores separately within the co-clusters avoids forcing unrelated profiles onto one axis.

Within blocks, antibiotic complexity increased across WHO AWaRe tiers (Spearman $\rho = 0.5895$, $p = 9.84 \times 10^{-5}$; 38 antibiotics with available scores). The association was weaker for global antibiotic complexity ($\rho = 0.4667$, $p = 0.0024$) and reached $\rho = 0.6244$ ($p = 8.49 \times 10^{-4}$) in the largest block. Taxa matching the ESKAPEE reference list had higher within-block pathogen fitness than other taxa: median standardised fitness was 1.044 for 16 matching taxa and -0.387 for 107 others (one-sided Mann–Whitney $p = 4.60 \times 10^{-6}$).

These comparisons provide an external consistency check because AWaRe and ESKAPEE were withheld from network construction and score calculation. The agreement does not make the quantities equivalent: AWaRe is an antibiotic-stewardship classification, ESKAPEE identifies a selected group of clinically important taxa, and Fitness–Complexity measures only the topology of RCA-positive links. Within-block scores are relative to each module and should not be interpreted as absolute resistance levels shared across modules.

Together, the structural and external-consistency results support a local use of Fitness–Complexity. The relevant comparison

is not whether one pathogen is universally “more resistant” than another, but how broad and selective its RCA-positive profile is relative to pathogens in the same network context.

Network exposure prospectively prioritises future signals

The rolling-origin back-test contained 1,839 eligible pathogen–antibiotic pair-years; 131 (7.12%) were RCA-positive in the following year. Antibiotic-network exposure achieved a pooled AUC of 0.7076, with origin-specific values from 0.6688 to 0.7703. Pathogen-network exposure achieved AUC 0.6283, while the antibiotic RCA-link-ubiquity baseline was close to random ranking (AUC 0.5123). An AUC of 0.7076 means that a randomly selected future RCA-positive event received a higher score than a randomly selected non-event in 70.76% of comparisons; it is not classification accuracy or an individual probability.

The event frequency increased across the exposure ranking. In sextiles of antibiotic-network exposure, 51 of 307 pairs (16.61%) became RCA-positive in the highest-exposure group, compared with 14 of 307 (4.56%) in the lowest. Both projected-network scores exceeded all 200 node-label permutation comparisons (empirical $p < 0.005$). Per-antibiotic AUCs were highest for ceftazidime, gentamicin, meropenem, cefepime and ceftriaxone, but these subgroup estimates contained only 5–9 events each and remain exploratory.

The signal was not confined to the next year. With all features frozen at 2019, antibiotic-network AUCs remained in the range 0.681–0.742 at horizons of one through five years. The prevalence baseline remained near chance (AUC 0.41–0.58). These comparisons include only pairs tested in the corresponding target year.

The conclusions were robust to the minimum-isolate threshold. Repeating the complete pipeline at $n_{\min} = 10, 30, 50$ preserved strong in-block nestedness and the ESKAPEE fitness separation. The default $n_{\min} = 30$ produced both the strongest within-block AWaRe association ($\rho = 0.590$, versus 0.547 and 0.442) and the highest rolling-origin AUC (0.708, versus 0.638 and 0.702), balancing low-count noise against loss of coverage.

Impact of the work

The framework adds an interpretable relational layer to routine AMR surveillance. Its path is explicit—counts $\rightarrow r_{pa} \rightarrow \text{RCA}_{pa} \rightarrow M \rightarrow$ relatedness network $\rightarrow$ exposure score—so each prioritised pair can be traced to the observations, existing links and network neighbours that generated its score. The ranking requires no fitted multivariable classifier and can be updated whenever new counts arrive. It complements pairwise summaries by identifying coherent resistance contexts, providing local comparison axes and focusing attention on future candidate signals.

The repository provides a deterministic end-to-end analysis script, reference mappings, output tables, publication figures and a static exploratory dashboard. The dashboard displays antibiotic-network positions, AWaRe labels, within-block scores, pathogen timelines and prioritisation scores. Its optional forward probabilities are generated by a separate one-variable calibration of antibiotic-network exposure; this display layer is not part of the parameter-free evaluation reported here and should not be interpreted as clinical risk estimation. The outputs could support surveillance review, additional testing or confirmatory analysis; they are not treatment recommendations.

The workflow can also be re-estimated for different periods or geographical subsets. Starting from isolate-level data, resistance fractions, RCA marginals, modules and networks must be recomputed within each selected country, region, continent or time window. Temporal re-estimation is demonstrated here. Geographical re-estimation is a deployment opportunity rather than a reported result, because the downstream table used for this analysis is pooled across countries.

Interpretation remains bounded by the surveillance universe. RCA is relative rather than absolute; ATLAS sampling and testing are not population-random; and the implemented zero combines sufficiently observed RCA-negative pairs with insufficiently observed pairs. The number of co-clusters is fixed at five, and the nestedness null is degree-corrected only in expectation. AWaRe and ESKAPEE provide external consistency rather than clinical validation. AUC measures ranking rather than calibrated individual risk, and a change in RCA status may reflect changing marginals or surveillance composition as well as biological change. The analysis does not establish causal acquisition, transmission, treatment failure or clinical outcome. Within these limits, it provides a reproducible and inspectable route from routine surveillance counts to structural understanding and future-priority signals.

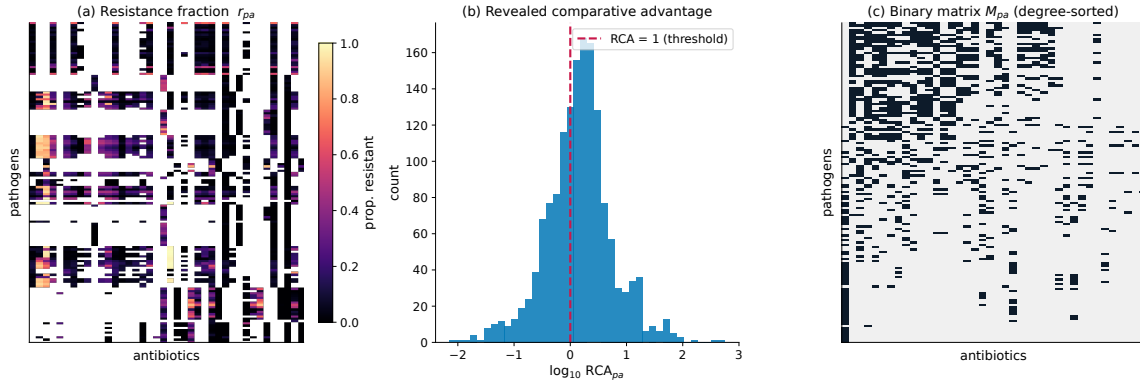


Figure 1: From surveillance counts to the bipartite network. Pooled resistance fractions (a) are normalised into RCA values (b) and thresholded at $RCA = 1$ to obtain M (c). RCA defines a comparative signal within the selected data and time window, not a clinical susceptibility category.

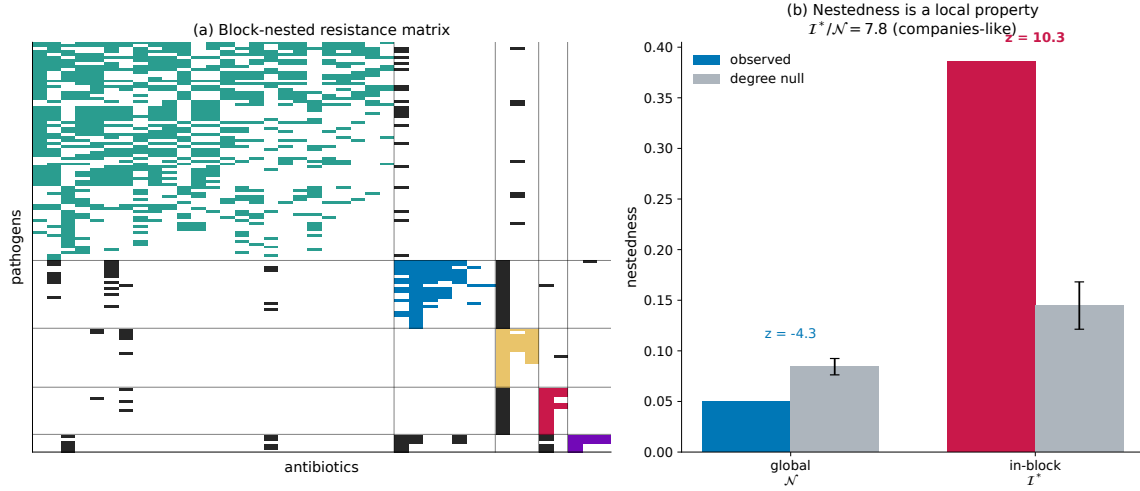


Figure 2: Global and local organisation of RCA-positive links. In (a), rows and columns are ordered by five spectral co-clusters and by degree within each cluster; coloured links lie within matched blocks and dark links connect different blocks. Panel (b) compares global nestedness $\mathcal{N}$ and in-block nestedness $\mathcal{I}^*$ with the degree-corrected Bernoulli null.

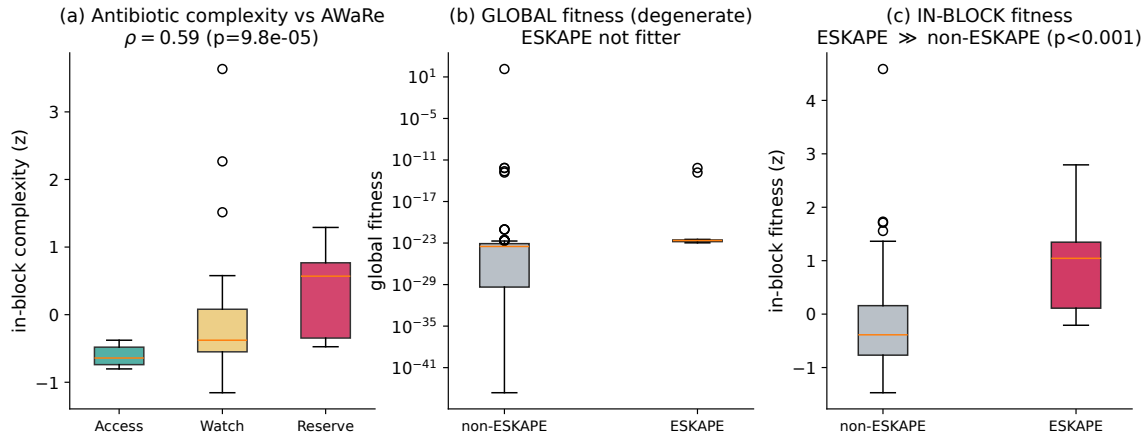


Figure 3: External consistency of the network rankings. Panel (a) compares within-block antibiotic complexity across WHO AWaRe categories. Panel (b) shows the collapse of the global Fitness–Complexity solution, whereas (c) compares within-block pathogen fitness between taxa in the ESKAPEE reference list and other taxa. Fitness is a network score, not evolutionary fitness.

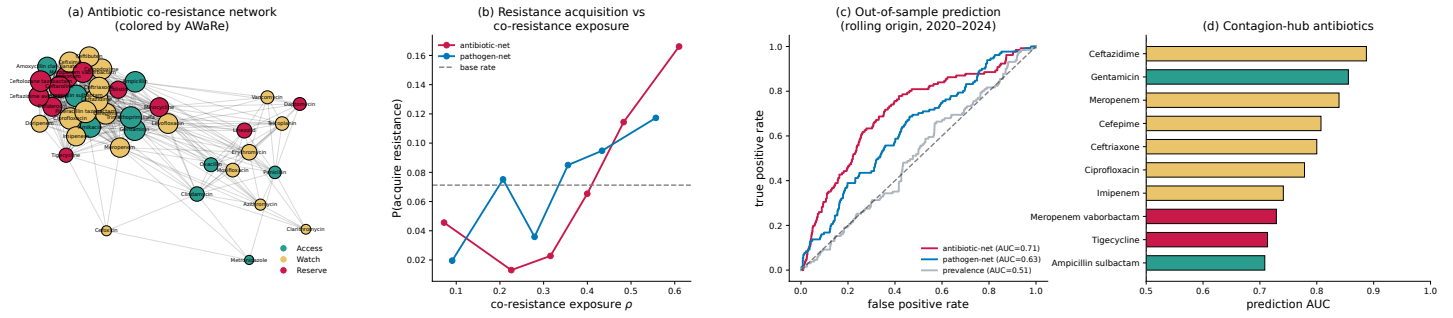


Figure 4: Prospective prioritisation. Panel (a) shows the antibiotic relatedness network; (b) shows next-year RCA-positive frequencies across exposure; (c) reports pooled prospective ROC curves; and (d) reports performance by antibiotic. Prediction uses raw co-occurrence weights; degree normalisation is used only for the network layout in (a). The event is a transition to RCA-positive status, not proof of biological acquisition.

References

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